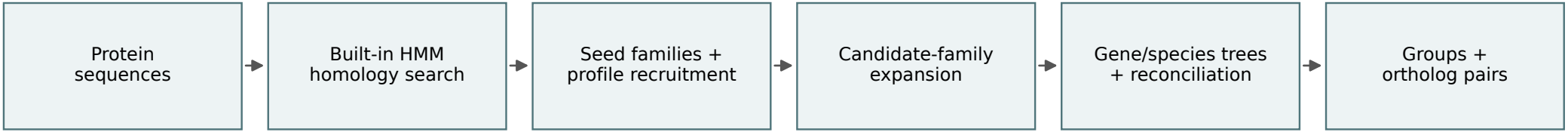
A HMM-centered inference with optional phylogenetic refinement



B Four separately controlled interventions

Sequence weighting

Relative residue weights; actual versus cluster effective N

Alignment-aware recruitment

Aligned residues, fragments and fusion/fission asymmetry

Local joint-tree rescue

Bounded neighboring families; reconciliation, not automatic union

Selective HMMER validation

Ambiguous candidates; separate measurement from reselection

Individual ablations > fixed advancement gates > justified interaction > full-vector reporting

HMM support nominates candidates; it does not alone establish orthology. Benchmark labels are evaluation-only.